

AI POWERED THREAT DETECTION SYSTEM

Complete Project Setup & Deployment Guide for External Users

1. Purpose of This Setup Guide

This document provides step-by-step instructions to download, configure, run, and troubleshoot the **AI Powered Threat Detection System** on any standard Windows, macOS, or Linux computer. It is designed so that a student, evaluator, external reviewer, or teammate receiving the project zip folder via Google Drive can run the complete application seamlessly without prior technical knowledge.

2. System Requirements

Resource / Requirement	Minimum Specs	Recommended Specs
Operating System	Windows 10/11, macOS 11+, or Ubuntu 20.04+	Windows 11 or Ubuntu 22.04 LTS
Processor (CPU)	Dual-Core 2.0 GHz	Quad-Core 2.5 GHz or higher
System Memory (RAM)	4 GB RAM	8 GB RAM or higher
Disk Storage	500 MB free space	1 GB free SSD space
Python Environment	Python 3.9, 3.10, 3.11, or 3.12	Python 3.10 or 3.11
Web Browser	Chrome, Edge, Firefox, or Safari	Google Chrome (Latest)

3. Step-by-Step Project Setup Instructions

Step 1: Download & Extract Project Files from Google Drive

- Open the shared Google Drive folder link in your web browser.
- Click the **Download All** button (or right-click the zip archive ai_network_attack_detection.zip and select Download).
- Once downloaded, extract the zip file to a clean local directory (e.g. C:\Projects\ai_network_attack_detection on Windows or ~/projects/ai_network_attack_detection on macOS/Linux).

Step 2: Install Python (If Not Already Installed)

- Download Python 3.10+ from official website: <https://www.python.org/downloads/>
- **CRITICAL STEP (Windows):** During installation, check the box that says 'Add Python to PATH' before clicking Install Now.
- Verify installation by opening Terminal / PowerShell and running:

```
python --version
```

Step 3: Open Terminal & Navigate to Project Folder

- Open Command Prompt, PowerShell, or Terminal.
- Navigate to the extracted project folder using the cd command:

```
cd C:\Projects\ai_network_attack_detection
```

Step 4: Create & Activate Virtual Environment (Recommended)

- Creating a Python virtual environment ensures package dependencies do not conflict with existing software:

```
# On Windows: python -m venv venv venv\Scripts\activate # On macOS / Linux: python3 -m venv venv
source venv/bin/activate
```

Step 5: Install Project Dependencies

- Install all required libraries using pip:

```
pip install -r requirements.txt
```

Required packages installed automatically: Flask, scikit-learn, joblib, numpy, pandas, matplotlib, reportlab, python-pptx, scapy.

Step 6: Launch the Master Application

- Execute the single master entrypoint script:

```
python main.py
```

• The script performs auto-verification: It checks for pre-trained model weights (dl_flow_model.joblib and rf_arp_model.joblib). If weights are missing, it automatically generates synthetic training datasets and trains models in ~10 seconds.

Step 7: Access the SOC Analyst Web Console

- Open your web browser (Chrome recommended) and navigate to:

```
http://127.0.0.1:5000
```

• The live dashboard will load with continuous simulated network traffic feed, interactive Attack Vector Simulator, live AI predictions, threat score graph, and security events table.

4. Common Errors & Troubleshooting Solutions

Error Description	Root Cause	Resolution / Fix
'python' is not recognized as an internal command	Python was installed without adding Python executable to OS PATH variable.	Re-run Python installer, select 'Modify', and check 'Add Python to environment variables'. Or use py main.py.
ModuleNotFoundError: No module named 'flask'	Dependencies were not installed into active Python environment.	Ensure virtual environment is activated and run pip install -r requirements.txt.
OSError: [Errno 98] Address already in use: 5000	Port 5000 is occupied by another local server (e.g. AirPlay on macOS or existing Flask app).	Kill process using port 5000 or edit web/app.py to change port to port=5050.
FileNotFoundError: dl_flow_model.joblib missing	Model files deleted or corrupt.	Run python models/train_models.py to re-generate datasets and retrain weights automatically.